

FinTrust Week 2 Project Summary

1. What I Planned to Accomplish

My Week 2 goal was to move from the planning completed in Week 1 into practical analysis. I planned to assess and prepare the FinTrust customer and transaction datasets, use SQL to answer business questions, perform exploratory analysis with Python, and develop an initial Power BI dashboard.

2. What I Completed

I completed an Excel-based data quality assessment, SQL business analysis, Python exploratory data analysis, and an initial Power BI management dashboard.

The analysis used the FinTrust customer dataset containing 1,500 customer records and the transaction dataset containing 12,000 transaction records.

During the quality assessment, I checked record counts, column counts, missing values, duplicates, unusual values, potential outliers, and the relationship between the customer and transaction datasets.

3. Tools Used

Activity	Tool
Data quality assessment	Microsoft Excel
Business analysis	SQL / DuckDB
Exploratory analysis	Python, Pandas, NumPy
Visual analysis	Matplotlib, Seaborn
Notebook environment	Google Colab
Dashboard	Microsoft Power BI

4. Key Business Findings

Finding 1: Everyday customers generated the highest transaction activity.

Evidence: The Everyday segment recorded 5,644 transactions.

Business Meaning: This segment contributes a large proportion of overall transaction activity and is therefore important when FinTrust considers service reliability and customer experience.

Finding 2: Mobile App was the most-used transaction channel.

Evidence: Mobile App recorded 5,102 transactions.

Business Meaning: Customers rely heavily on FinTrust's mobile channel, making mobile service performance particularly important.

Finding 3: Most transactions were successful.

Evidence: 10,856 transactions were successful, representing approximately 90.47% of all transactions.

Business Meaning: Overall transaction completion is strong, although failed, reversed and pending transactions should still be monitored for possible operational issues.

Finding 4: Transfers and card purchases were the most frequent transaction types.

Evidence: Transfers recorded 3,549 transactions while card purchases recorded 3,033.

Business Meaning: These two activities represent a major part of customer transaction behaviour and may require greater operational attention.

Finding 5: 19.6% of transactions were flagged for risk review.

Evidence: 2,352 of the 12,000 transactions had Risk_Review_Flag = Yes.

Business Meaning: These transactions provide a useful group for further pattern analysis. The Risk Review Flag is a synthetic educational indicator and does not mean fraud occurred.

Testing and Evaluation

Work Completed	Test / Evaluation	Finding	Action / Improvement	Result
Excel data assessment	Checked record counts, missing values and duplicates	Customer data complete; minor missing transaction fields	Retained valid transactions with missing Device_Type/Location	Dataset remained suitable for analysis
Customer/transaction relationship	Validated Customer_ID relationship	All transaction customer IDs matched valid customers	Created one-to-many relationship	Datasets could be analysed together
SQL analysis	Compared query outputs with previously profiled totals	Results were consistent with dataset totals	Retained queries and interpretations	8 business questions completed
Python EDA	Checked chart results against SQL counts	Segment, channel and status results were consistent	Used results in business findings	EDA supported dashboard development
Power BI model	Tested relationship and KPI values	Relationship correctly filtered customer and transaction data	Used active one-to-many relationship	Dashboard visuals responded correctly
Power BI trend	Initial trend chart counted individual timestamps	Monthly trend initially displayed incorrectly	Created a month-level date field	Monthly Jan–Mar trend displayed correctly

Important Decisions

The missing Device_Type and Location values were retained because the affected records still contained valid transaction information. Removing them would have caused unnecessary loss of otherwise usable transaction data.

High-value transactions were also retained because there was no evidence that they were data-entry errors.

For risk analysis, Risk_Review_Flag was treated only as an educational review indicator and not as evidence of fraud, as required by the project documentation.

Limitations

The FinTrust dataset is synthetic and does not represent real banking customers or transactions. A small amount of missing information exists in Device_Type and Location, which may affect analyses involving those fields.

The transaction data covers only January to March 2026, so longer-term behaviour or seasonal trends cannot be determined from the available period.

Changes From Week 1

Week 1 focused mainly on understanding the problem, profiling the data and planning the analysis.

In Week 2, those plans were implemented using Excel, SQL, Python and Power BI. The analytical questions developed in Week 1 were used as the basis for the SQL and exploratory analysis.

Week 3 Focus

Week 3 should focus on refining the dashboard, investigating the strongest findings in greater depth, validating KPI calculations, improving presentation quality, and developing clearer recommendations from the analysis.